

SURV/SURVMETH 720
Total Survey Error & Data Quality 1

Repairs for Undercoverage

Class #3
September 15, 2025

Mappings between Frame and Target Population

Frame	Target Population	
F	T	Correspondence
.	T	Undercoverage
F	.	Foreign elements
F, F	T	Multiplicity (many-to-one)
F	T, T	Clustering (one-to-many)

About “overcoverage”

Different researchers use the term *overcoverage* inconsistently, so it is important to consider whether over-coverage in a given sample frame is caused by ineligible units, multiple records, or both.

[Encyclopedia of Survey Research Methods](#)

Measuring Undercoverage

Most of the time the researcher has limited means of measuring coverage error

- Both the noncoverage rate and difference between the noncovered and covered populations are unknown

1. One strategy is to compare data from frame to external data (assumed to be better), e.g., Census data or data from another frame

Measuring Undercoverage...

2. Another strategy is analyze data from large-scale survey where membership on a frame (that was not used by the survey) is measured

- e.g., Phone access measured in a mail survey using ABS.

3. Coverage check studies

- National Survey of Black Americans
- Idea was to reduce screening cost
 - Screen only in some homes
 - Ask if other blacks lived in the area
- Coverage check: Visit all HHs in an area
 - Screening question in each HH
 - Also ask information on other blacks
 - Yields rate of non-coverage of the cheaper method. Also answers: are we missing out on certain types of people ?

But coverage rate != coverage bias

Amaya et al. (2021) assess ABS for RECS 2015

Step 1:.. Assess coverage rate within each sampled segment by comparing fieldable residential addresses to the ACS HU count.

Step 2: Use one of three frames: ABS only (if est. cov. rate $> 90\%$), enhanced listing (if $56\% < \text{ECR} \leq 90\%$), field listing (if $\text{ECR} \leq 56\%$).

- Observed coverage rates are 100%, 96%, 25%

Amaya et al. (2021)...

Step 3: Build a population based on the RECS sample.

Step 4: Fit a logistic model: covered by original ABS (yes, no) ~ address and block group information from ACS

Step 5: Construct 100 frames with coverages ranging from 1 – 100% by dropping bottom x% of the pop. based on propensity.

Amaya et al. (2021)...

Step 6: Draw 1000 samples of 1000 addresses from each of the 100 frames.

Step 7: We know the pop. value and the estimate for each sample. Calculate bias and relative bias for the 12 variables of interest: #bedrooms, R's educ. and age, Fuel type, HH size, HU size...

Amaya et al. (2021)...

- Results:
 - Low risk of coverage bias if rate $> 75\%$: education, HH size, year built
 - But much riskier for fuel type, race, census region, and urbanicity once coverage rate $< 90\%$

Idea for 721:

First, researchers may live with the risk of producing biased estimates. If the purpose of the research is not to produce prevalence estimates but, instead, to build models or review the relationship among variables, it is possible these analyses will be unbiased. Second, one may take the opposite

Methods for reducing coverage error

1. Redefine the target population
2. Enhance the sample frame
3. Statistical adjustments

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1. Redefine the target population

Define the target population in a way that is compatible with the sample frame

- Redefine scope of the target population
 - Adults living in the U.S. -> civilian population living in the lower 48 states and District of Columbia
 - Head and spinal cord injury victims -> head and spinal cord injury victims who required hospitalization
 - Credit card users -> credit card users banking with the University of Michigan Credit Union
- Redefine the population units
 - X users -> tweets
 - visitors (to a museum, business, web site) -> visits

1. Redefine the target population

- Problem: Relevance
- Tradeoff between two dimensions of data quality: accuracy and relevance
 - Effective surveys have a target population that is well represented on the frame AND of interest to data users

Group activity: https://bit.ly/720_2025

[Class3](#)

Methods for reducing coverage error

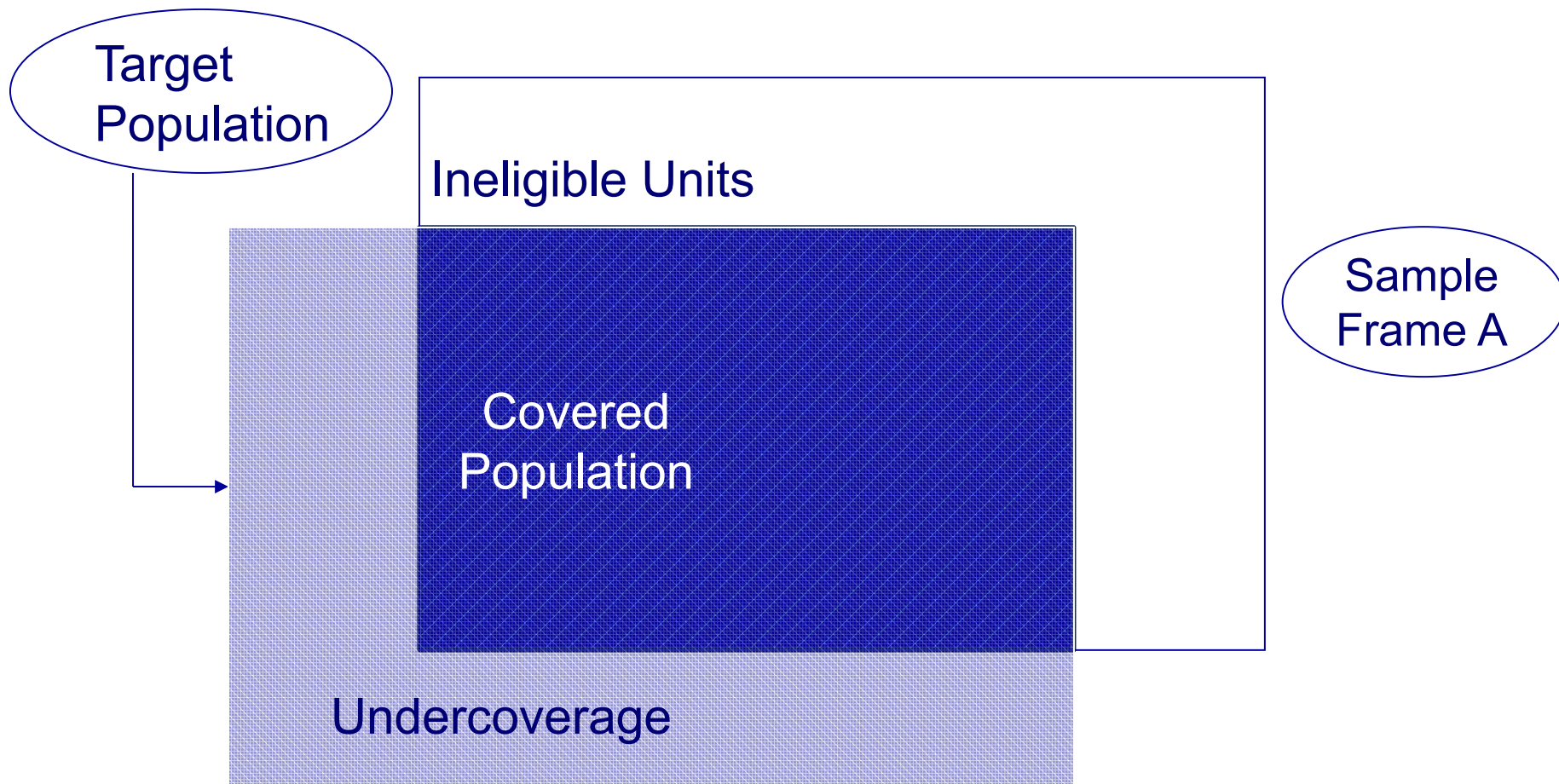
1. Redefine the target population
2. Enhance the sample frame
3. Statistical adjustments

Methods for enhancing the sample frame

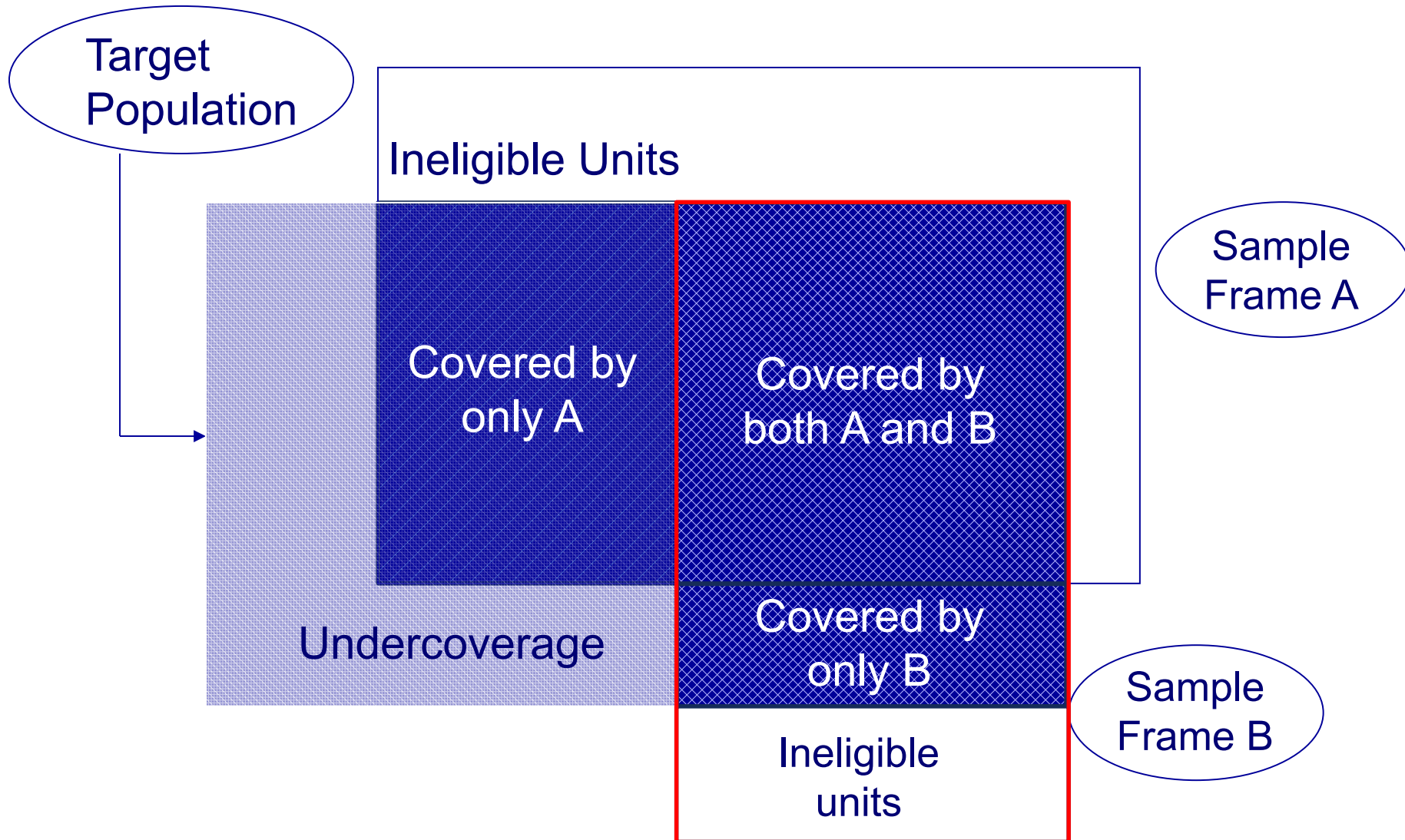
1. Multiple frames
2. Linking methods
3. Network sampling

1. Multiple frames

- Add one more sample frames
- Determine whether each sampled unit is a member of another frame
 - Example: In Dual-Frame RDD, need to know whether each respondent called on landline also has a cell phone (and vice versa)
- Compute selection probabilities based on information about frame membership
 - Example: Individuals with both landline and cell phones have higher likelihood of selection



Dual-Frame representation



Application: Improving coverage of homeless persons. Iachan and Dennis (1993)

- Target population ?
- Characteristics ?
- Location-time sampling.
- Adjust for multiplicity.

Iachan and Dennis (1993)....

Four frames

- Shelters
- Soup kitchens
- Street area frame
- Encampments

Does all this matter?

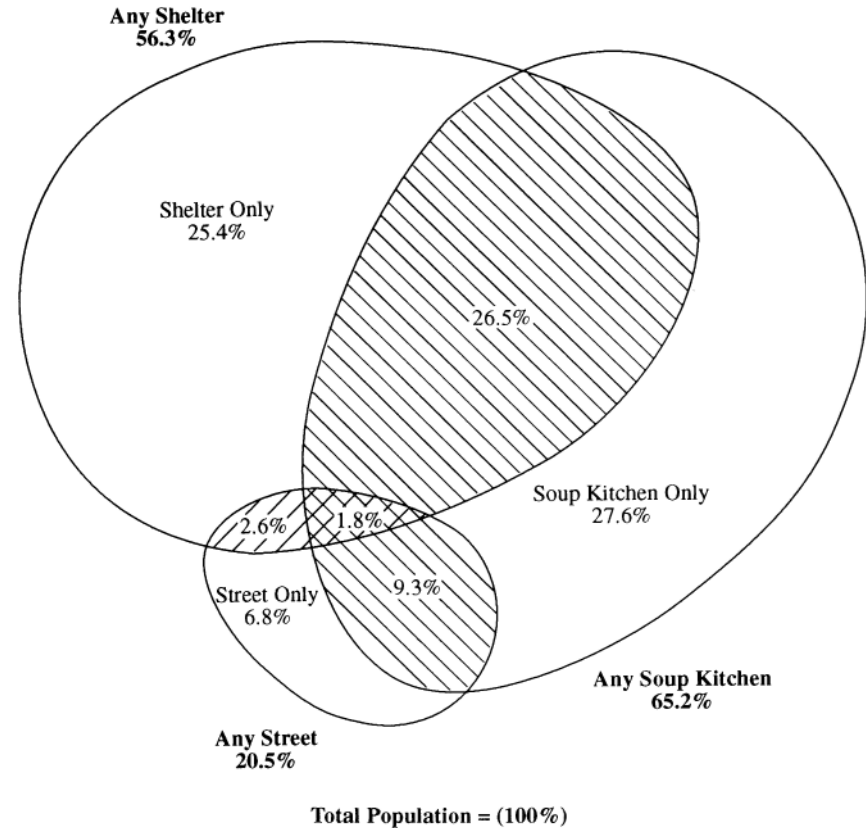
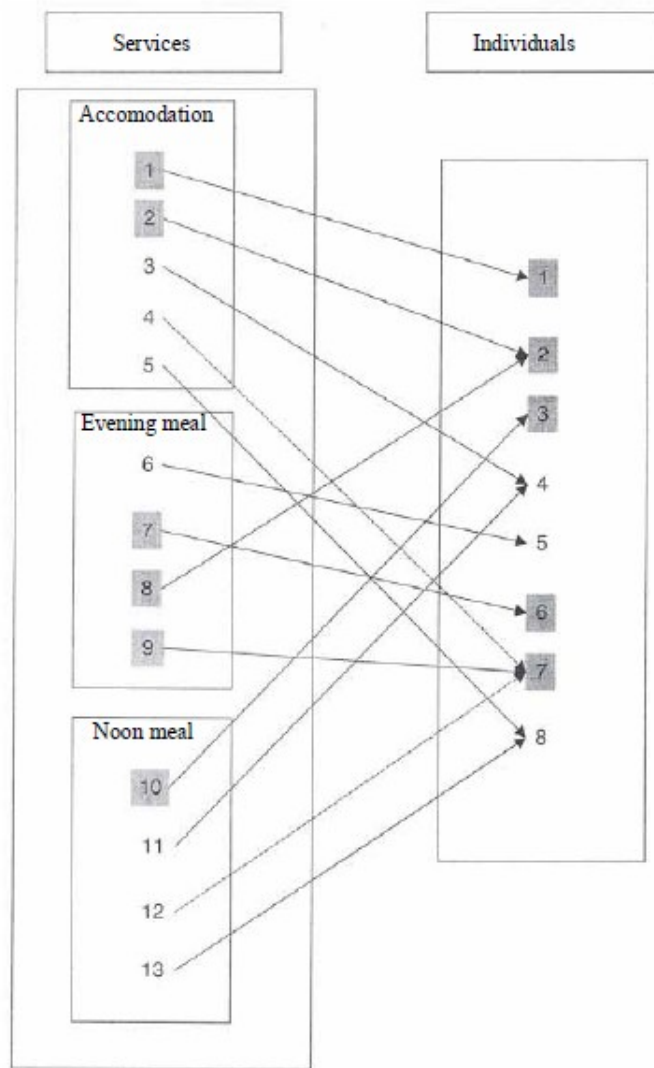


Fig. 1. Size and 24-hour overlap of the three subpopulations of homeless people in the DC MSA

The total size of the homeless and transient population on an average day in February through June 1991. Failure to take even the 24-hour overlap into account would have increased the estimate by 42% and partially explains why local estimates vary so much among researchers, practitioners, and advocates of this population.

$$\tilde{w}_1 = w_1, \tilde{w}_2 = \frac{1}{2}(w_2 + w_8), \tilde{w}_3 = w_{10}, \tilde{w}_6 = w_7, \tilde{w}_7 = \frac{1}{3}w_9.$$



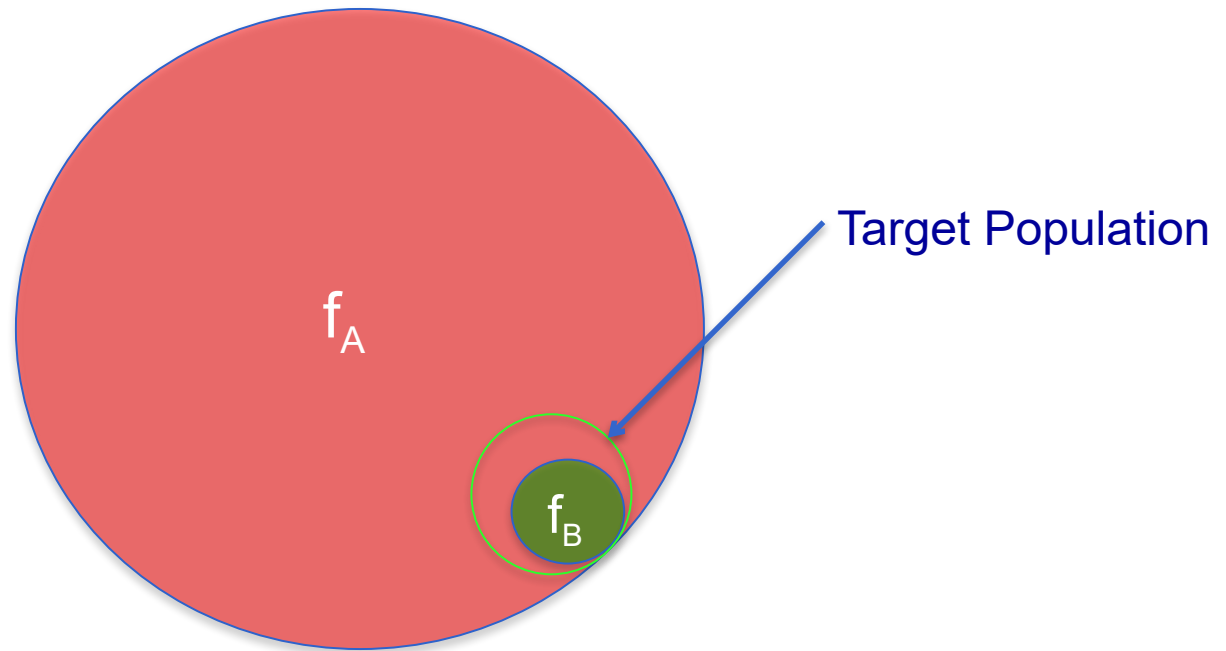
Weight sharing method

*Ardilly and Le Blanc 2001
(has a nice section on defining the population)*

An individual's weight is equal to the sum of the weights of the services that were used to "catch him", divided by the number of links with the survey frame.

On multiple frames: also see Hyman et al (2021) for reducing undercoverage in an agricultural survey

But can also use multiple frames to get more efficient.



Application:

Pew Survey of Muslims (2017)

- Called cellphones in addition to landlines
- Added group of phone numbers that had been flagged by third-party data vendors as likely to belong to a Muslim

Number of interviews in each frame, stratum

National random sample of cellphones	Total screener interviews (including both Muslims and screen-outs)	Total Muslim interviews	Muslim incidence rate
Very high Muslim density stratum	4,913	149	1 in 33
High Muslim density stratum	2,653	43	1 in 62
Medium Muslim density stratum	6,895	90	1 in 77
Low Muslim density stratum	5,656	54	1 in 105
Very low Muslim density stratum	9,607	63	1 in 152
Lowest Muslim density stratum	Excluded	Excluded	n/a
Flagged cellphone stratum	1,438	385	1 in 4

Complications with using multiple frames

- Determining selection probabilities
 - Need information about frame membership
- Determining the proportion of the sample to select from each frame
 - Requires balancing cost considerations and sampling error considerations
- Often requires multiple data collection procedures

Methods for enhancing the sample frame

1. Multiple frames
2. Linking methods
3. Network sampling

2. Linking Methods

- Adding missed units to an existing frame by linking them to selected units
 - Requires a linking rule

Half-Open Interval (HOI)

- Adds housing units to an address frame by searching for new or missed units in the interval between the selected unit and next unit in the frame
 - Why “Half” ?
 - Used for in-person interviewing; can be applied to either area sampling or ABS frames
 - Procedure is carried out by interviewers during data collection

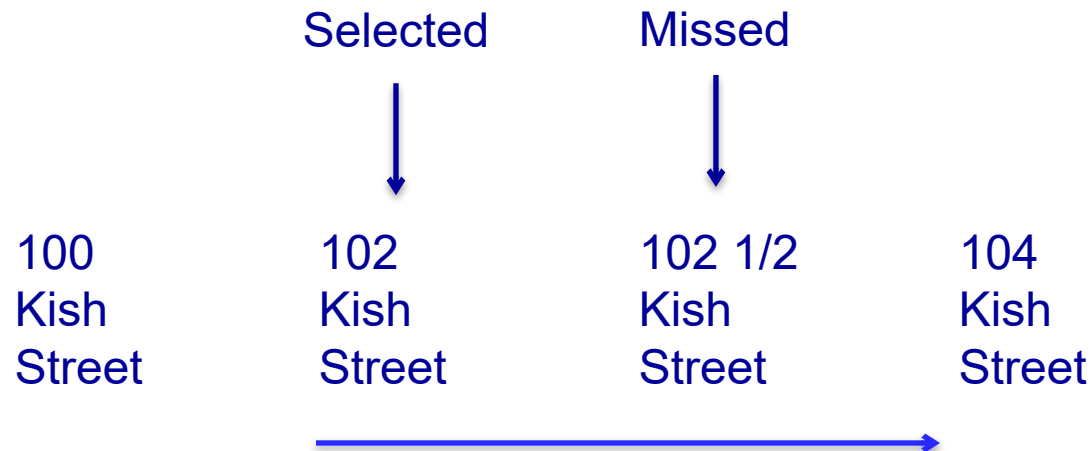
HOI Example

Housing List:

100 Kish St

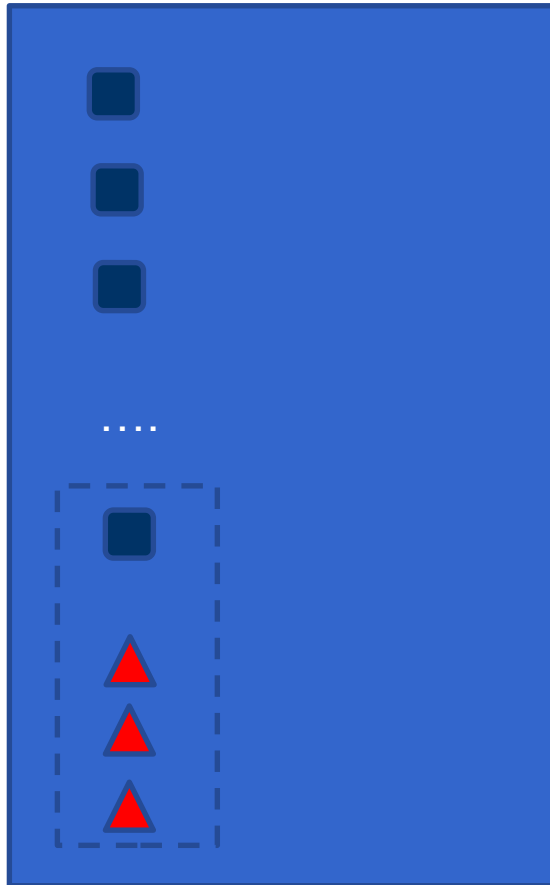
102 Kish St (selected)

104 Kish St.



■ → On the frame

▲ → Missing



If drawing sample of employees for survey but sample was drawn from an out-of-date list. Can create linking rule whereby all new employees are linked to the last one on list. If last one is selected, then efforts are made to identify all new employees.

Benefits and Complications of Linking Methods

- Benefits:
 - Increases coverage
- Complications:
 - Extra effort required (expensive)
 - Increases uncertainty of sample size (can subsample)
 - Subject to listing errors (Eckman and O'Muircheartaigh (2011))

Methods for enhancing the sample frame

1. Multiple frames
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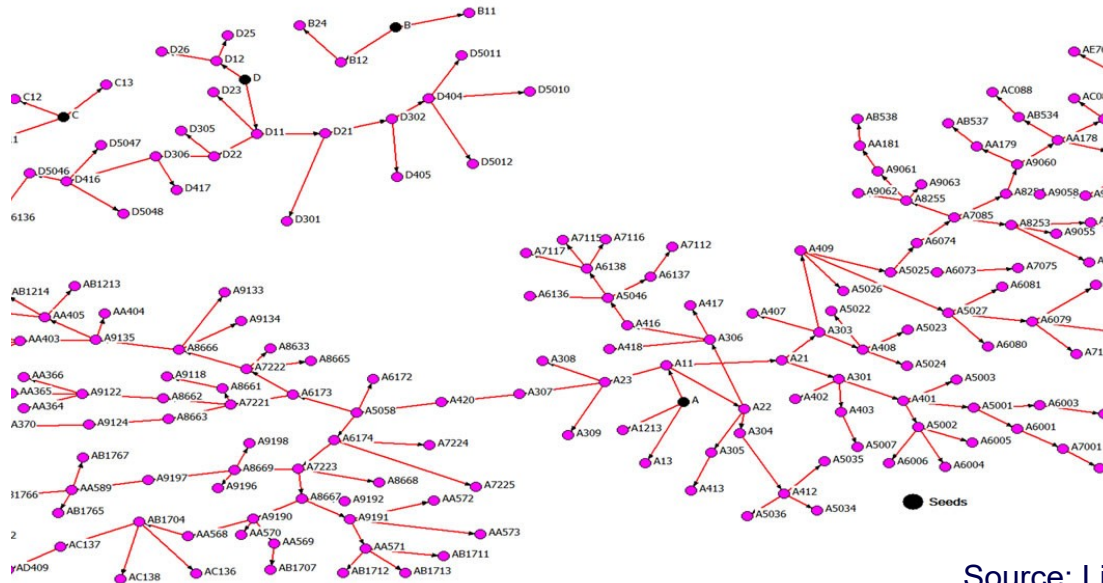
Hard-to-reach populations

Network Sampling

- What do we mean by “hard-to-reach”?
- Multiplicity
 - Respondents report about events among those in their personal networks (i.e., acquaintances, friends, extended family)
- Snowball sampling (non-probability)
 - Initial recruits in convenience sample refer to other members in the population.

Respondent-driven sampling (RDS)

- Effort to convert snowball sampling into a probability sampling method
- Recruit 'seeds' who then recruit others in their network



Source: Liu, Li, Ha, and Li (2012)

RDS...

- Must be possible to trace who recruited whom
- Should collect information concerning the size of each person's network;
- Must have a maximum number of possible recruitments for a person
- Recruiter and recruit must have known each other prior to recruitment, each being a member of the other's social network.
- But not fully straightforward...

RDS – Potential Problems

- Can make strong assumptions about random selection, e.g., choice of network member is random.
- Recruitment...

...see Marpsata and Razafindratsima (2010).

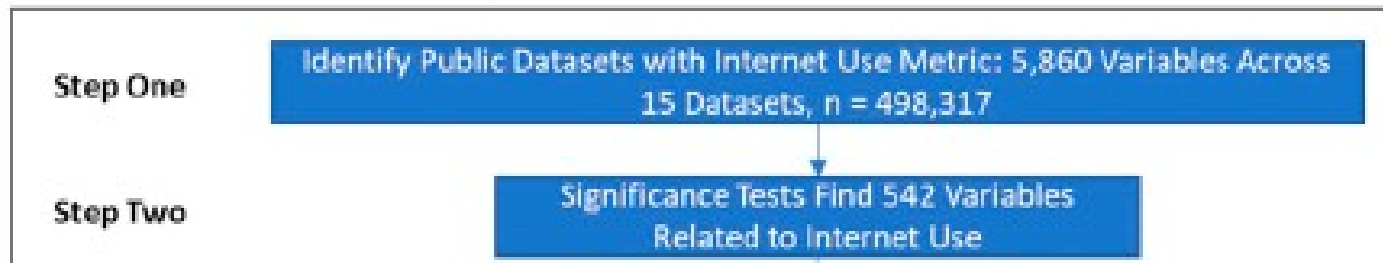
Methods for reducing coverage error

- Redefine the target population
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Statistical adjustments

- General idea: Develop weights/calibrate to account for the non-covered population
- Example: Internet surveys leave out members with no internet access. Can weighting help bridge possible differences?

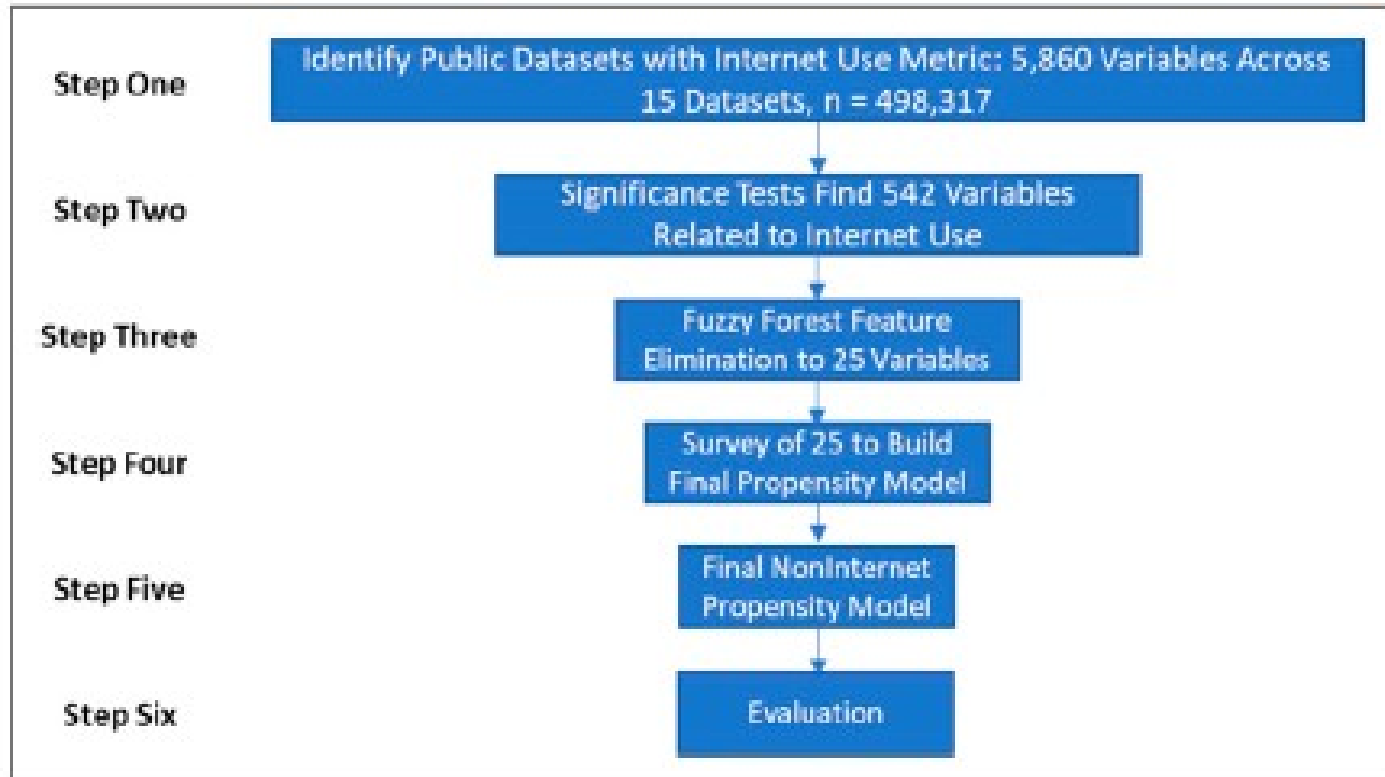
....Reducing Coverage Bias in Internet Surveys (Dutwin and Buskirk 2023)



- High internet penetration does not mean coverage bias is negligible.

authors reported their results. There are 122 variables for which the estimate for the internet population is at least double that for the non-internet population, or vice versa. To name just a few: own a computer (82% of the internet population vs. 20% for the non-internet population); went to see a movie in the past year (75% vs. 32%); own stocks (71% vs. 28%); became a U.S. citizen before 1960 (4% vs. 36%); importance to choose products politically (not important = 5% vs. 25%); and has given to charity (77% vs. 42%). Given these wide gaps, even with the non-internet population under 10 percent of the total population, a survey with only Internet respondents would experience bias of as much as 9% on these measures, and 105 variables would exhibit a bias of at least 5 percent.

....Reducing Coverage Bias in Internet Surveys (Dutwin and Buskirk 2023)



....Reducing Coverage Bias in Internet Surveys (Dutwin and Buskirk 2023)

Table 5. Random Forest Importance Measures, Final Omnibus Dataset.

Variable
Age
Go to movies past year
Know nanotechnology
Employment
Marital status
Education
Income
Know...Had a Baby...Year
Number of adults
Contacted politician...year

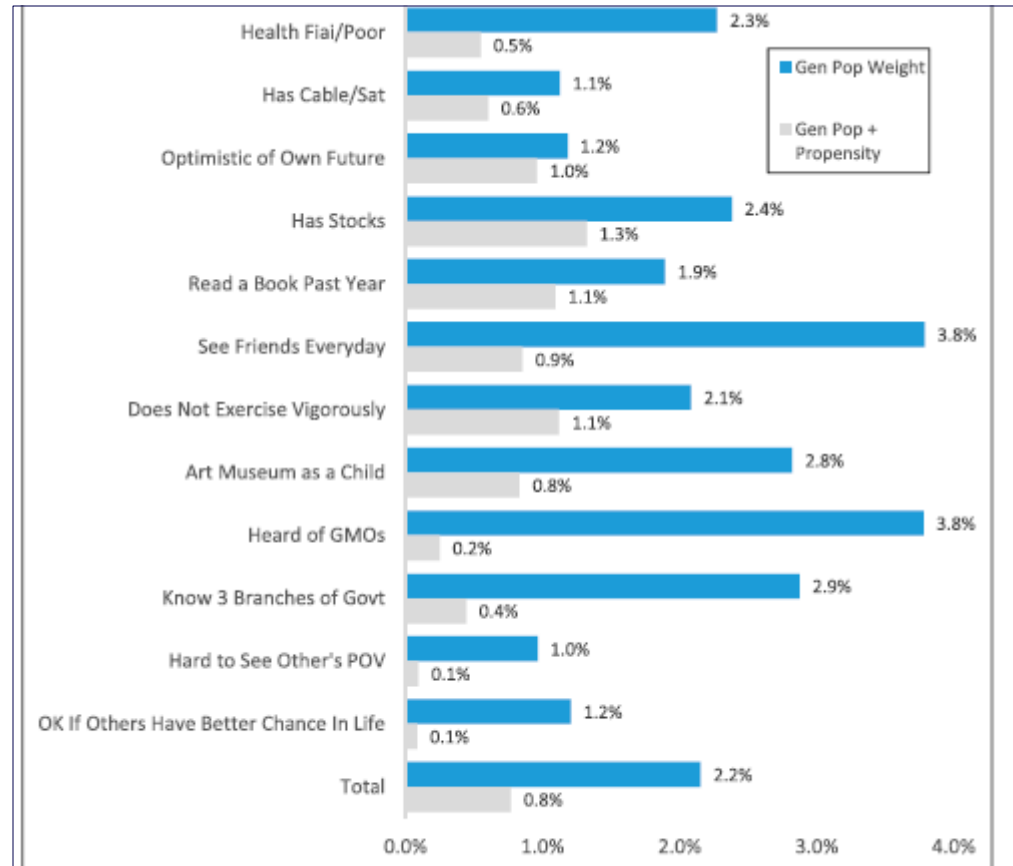


Figure 3. Mean absolute biases: Internet population, gen pop wgted versus gen pop + propensity wgted.

Also see Dever et al. (2008) for an assessment of weighting to reduce coverage bias

- Consider a Web survey of the general population that omits those without Internet access. Those with access to the Internet differ from those without access on the survey variables. Do you see this as a "coverage" issue or "nonresponse" issue?
 - Coverage
 - Nonresponse

Providing Way for Sample Members to Access Survey

- Mail/phone option in Web survey
- Translating questionnaire into multiple languages
- Providing Internet-enabled devices in Web survey
 - Expensive, training and tech support may be required (thus, more practical for panels than one-time surveys)

Providing Internet-Enabled Devices

Examples

- LISS (Longitudinal Internet Studies for Social Sciences)
 - Selected households from Dutch population registry
 - Provided computer to non-Internet households
 - Eckman (2016): Exclusion of non-internet households in analyses of LISS data would bias a large number of estimates of means (but few of the estimates of regression coefficients)

Next Week's Class

- Note: Read all the required readings
- Nonresponse
- Questions?